

Polynomials (多項式) (24/8/2022)

A function (函數) of the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where n is a positive integer, is called a polynomial in one variable (變元). The numbers $a_n, a_{n-1}, \dots, a_1, a_0$ are called the coefficients (係數) of the polynomial.

Further, if $a_n \neq 0$, then ' n ' is the degree (次) of the polynomial, denoted by $\deg f$.

Two polynomials $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ and $g(x) = b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0$ are equal if and only if $m = n$ and all the corresponding coefficients are equal, and we write $f(x) \equiv g(x)$.

Operations on polynomials (多項式的運算)

Define 1. $f(x) \pm g(x) = (a_n \pm b_n) x^n + (a_{n-1} \pm b_{n-1}) x^{n-1} + \dots + (a_1 \pm b_1) x + (a_0 \pm b_0)$

2. $k(f(x)) = k a_n x^n + k a_{n-1} x^{n-1} + \dots + k a_1 x + k a_0$

3. $f(x) \cdot g(x) = a_n b_m x^{n+m} + (a_{n-1} b_m + a_n b_{m-1}) x^{n+m-1} + \dots + (a_1 b_0 + b_1 a_0) x + a_0 b_0$

4. $f^m(x) = (a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0)^m$, where m is a positive integer.

Example 1: Let $f(x) = x^4 + x^3 - 3x^2 - x + 2$. Prove that for any positive integer m , the polynomial $f^m(x)$ has always one negative coefficient.

Division Algorithm (多項式的除法)

Suppose $\deg f(x) \geq \deg g(x)$, when $f(x)$ is divided by $g(x)$, we obtain the quotient $Q(x)$ and the remainder $R(x)$, where $\deg R(x) < \deg g(x)$. That is, $f(x) = g(x)Q(x) + R(x)$

Example 2: Divide $2x^4 - 11x^3 - 23x^2 + 20x + 5$ by $x^2 - 7x + 1$.

Answer: The quotient is $2x^2 - 3x - 4$, and the remainder is $-11x + 9$.

Example 3: Divide $3x^4 + 7x^3 - 15x - 20$ by $x + 2$.

Answer: The quotient is $3x^3 + x^2 - 2x - 11$ and the remainder is 2

Example 4: Divide $15x^3 - 8x^2 - 2x + 4$ by $3x - 1$.

Answer: The quotient is $5x^2 - x - 1$ and the remainder is 3.

Example 5: Find the remainder when $x^{60} - 2x^{25} + 1$ is divided by $x^2 - 1$.

Answer: Remainder = $-2x + 2$.

Remainder Theorem (餘數定理)

Theorem: The remainder when $f(x)$ is divided by $(x-b)$ is $f(b)$.

Proof: Suppose $f(x) = (x-b)Q(x) + R$, where $\deg(R) = 0$.

Putting $x = b$ gives $f(b) = R$.

Factor Theorem (因式定理)

Given the polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, $a_n \neq 0$. The binomial $(x-c)$ is a factor of $f(x)$ if and only if $f(c) = 0$.

Example 6: Factorize $f(x) = 9x^4 - 3x^3 + 7x^2 - 3x - 2$.

Answer: $f(x) = (3x+1)(3x-2)(x^2+1)$

Example 7: Prove that $x(x+1)(2x+1)$ is a factor of $(x+1)^{2m} - x^{2m} - 2x - 1$.

Example 8: Given that a, b, c, d are positive integers, prove that $f(x) = x^{4a} + x^{4b+1} + x^{4c+2} + x^{4d+3}$ is divisible by $x^3 + x^2 + x + 1$.

Example 9: Prove that $(a+b)(b+c)(c+a) + abc = (a+b+c)(ab + bc + ca)$

Example 10: Prove that $(a+b+c)^3 - a^3 - b^3 - c^3 = 3(a+b)(b+c)(c+a)$

Factorization of Homogeneous Polynomials (齊次多項式)

Example 11: Factorize $P(a, b, c) = a^3 + b^3 + c^3 - 3abc$

Solution: $P(a, b, c) = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Example 12: Prove that $(a+b)(b+c)(c+a) + abc = (a+b+c)(ab + bc + ca)$

Solution: $(a+b)(b+c)(c+a) + abc = (a+b+c)(ab + bc + ca)$

Example 13: Factorize $F(a, b, c) = \sum (b-c)(b+c)^3$.

Solution: $F = -2(a-b)(b-c)(c-a)(a+b+c)$.

Example 14: Factorize $F(a, b, c) = a^3(b^2-c^2) + b^3(c^2-a^2) + c^3(a^2-b^2)$

Solution: $F(a, b, c) = -(a-b)(b-c)(c-a)(ab+bc+ca)$

Example 15: Factorize $F(a, b, c) = \sum_{abc} (a+b)^2(b+c)^2(b-c)$

Solution: $F(a, b, c) = -(a-b)(b-c)(c-a)(a^2+b^2+c^2+ab+bc+ca)$

Example 16: Factorize $F(a, b, c, x) = \sum_{abc} (a-b)^3(x-c)^2$

Solution: $F(a, b, c, x) = (a-b)(b-c)(c-a)(3x^2 - 2x(a+b+c) + ab+bc+ca)$

Example 17: (17th IMO 1975 No. 6) Find all polynomials P , in two variables, with the following properties:

- (i) for a positive integer n and all real t, x, y , $P(tx, ty) = t^n P(x, y)$
(that is, P is homogeneous of degree n),
- (ii) for all real a, b, c , $P(b+c, a) + P(c+a, b) + P(a+b, c) = 0$
- (iii) $P(1, 0) = 1$.

Solution: $P(x, y) = (x+y)^{n-1}(x-2y)$, for $n \geq 1$.

Example 18: (45th IMO 2014 No. 2) Find all polynomials f with real coefficients given that for all real numbers a, b, c such that $ab+bc+ca = 0$ we have the following relationship $f(a-b) + f(b-c) + f(c-a) = 2f(a+b+c)$.

Solution: $f(x) = p x^4 + q x^2$ satisfies the condition for any p, q .

Eisenstein Criterion (愛森斯坦判別法)

(Named after Gotthold Eisenstein, 1823-1852, a German Mathematician)

If $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, a_2, \dots, a_n$ are all integers, and there exists a prime number p such that (i) $p|a_i$ for $i = 0, 1, 2, \dots, n-1$, (ii) $p \nmid a_n$, (iii) $p^2 \nmid a_0$, then $f(x)$ is not reducible over the set of rational numbers. (不可約多項式).

Proof: Assume that there exist $g(x)$ and $h(x)$ such that $f(x) = g(x)h(x)$, and

$$g(x) = b_r x^r + b_{r-1} x^{r-1} + \dots + b_1 x + b_0, h(x) = c_s x^s + c_{s-1} x^{s-1} + \dots + c_1 x + c_0, r, s \in \mathbb{N}^+$$

We have $a_0 = b_0 c_0$. Since $p|a_0$ and $p^2 \nmid a_0$, hence w.l.o.g. we can assume that $p|b_0$ and $p \nmid c_0$.

On the other hand, since $p \nmid a_n$, and $a_n = b_r c_s$, hence $p \nmid b_r$.

Now consider $b_0, b_1, \dots, b_{r-1}, b_r$, starting from b_0 , we have the coefficients divisible by p , until we come to a certain b_k , where $0 < k < r$.

(i.e., $p|b_0, p|b_1, p|b_2, \dots, p|b_{k-1}, p \nmid b_k, p \nmid b_{k+1}, \dots, p \nmid b_r$)

Next consider $a_k = b_k c_0 + b_{k-1} c_1 + \dots + b_0 c_k$

Every term on RHS is a multiple of p except the first term, hence a_k is not divisible by p .

This is contradictory to condition (i).

Hence the assumption is wrong, and $f(x)$ is irreducible.

Example 19: The polynomial $f(x) = 3x^4 - 10x^3 + 30x + 15$ is irreducible.

Example 20: Prove that the polynomial $f(x) = 2x^4 + 9x^2 + 6$ is irreducible. (Take $p=3$)

Example 21: The polynomial $f(x) = x^7 - 36x^2 + 15$ is irreducible. (Take $p = 3$)

Example 22: Prove that the polynomial $f(x) = x^2 - k$, where k is a prime number, is irreducible.

Note that this is a sufficient but not necessary condition.

For example, $f(x) = x^2 + 1$ is irreducible, but no suitable p can be found.

Example 23: Prove that the polynomial $f(x) = x^2 + x + 2$ is irreducible.

(Hint: Put $x = y+3$ to change the polynomial to $y^2 + 7y + 14$, and take $p=7$.)

Example 24: Prove that the polynomial $f(x) = x^4 + 1$ is irreducible.

Example 25 If p is a prime number, prove that the polynomial $f(x) = x^{p-1} + x^{p-2} + x^{p-3} + \dots + x + 1$ is irreducible.

Euclidean Algorithm(輾轉相除)

If $f(x)$ and $g(x)$ are polynomials of degrees n and m , both greater than 1. Suppose there exists a polynomial $h(x)$, unique apart from a constant factor, such that $h(x)$ is the Greatest Common Divisor of $f(x)$ and $g(x)$, then $h(x)$ can be found using the Method of Euclidean Algorithm, as follows:

Method: Given the polynomials $f(x)$ and $g(x)$, with all common constant factors removed beforehand.

Suppose $\deg f(x) \geq \deg g(x)$, then $f(x) = g(x) q_1(x) + r_1(x)$ -----(1)

$$g(x) = r_1(x) q_2(x) + r_2(x)$$

$$r_1(x) = r_2(x) q_3(x) + r_3(x)$$

$$r_2(x) = r_3(x) q_4(x) + r_4(x)$$

.....

$$r_{n-2}(x) = r_{n-1}(x) q_n(x) + r_n(x)$$

$$r_{n-1}(x) = r_n(x) q_{n+1}(x)$$

Finally, $r_n(x)$ is the G.C.D. of $f(x)$ and $g(x)$.

Proof: Consider (1). Let $d(x)$ be the Greatest Common Divisor (G.C.D.) of $f(x)$ and $g(x)$ and $d'(x)$ be the G.C.D. of $g(x)$ and $r_1(x)$.

Since $d'(x)$ be the G.C.D. of $g(x)$ and $r_1(x)$, then it is a factor of $f(x)$, and hence $d(x)$. Similarly by rewriting in the form $r_1(x) = g(x) q_1(x) - f(x)$, we see that since $d(x)$ is a factor of both $f(x)$ and $g(x)$, this means that $d(x)$ is a factor of $r_1(x)$, and hence a common factor of $r_1(x)$ and $g(x)$ and hence a factor of $d'(x)$.

From above we see that $d(x)$ and $d'(x)$ are factors of each other and they differ by at most a constant.

As common constant factors have been removed beforehand, therefore the G.C.D. of $f(x)$ and $g(x)$ is the same as the G.C.D. of $g(x)$ and $r_1(x)$.

Repeating the above procedure, we can say that the G.C.D. of $f(x)$ and $g(x)$, $g(x)$ and $r_1(x)$, $r_1(x)$ and $r_2(x)$, $r_2(x)$ and $r_3(x)$, ..., $r_{n-1}(x)$ and $r_n(x)$ are all the same.

Now $r_n(x)$ is a factor of $r_{n-1}(x)$.

Hence $r_n(x)$ is the G.C.D. of $r_{n-1}(x)$ and $r_n(x)$, and hence $f(x)$ and $g(x)$.

Example 26: Find the G.C.D. of $f(x) = x^5 - x^4 + 8x^2 - 8x$ and $g(x) = x^3 + 4x^2 - 8x + 24$.

Solution: The G.C.D. is $x^2 - 2x + 4$

Example 27: Find the G.C.D. of $f(x) = 3x^4 - 11x^3 + 15x^2 - 6x$ and $g(x) = 2x^3 - 12x^2 + 24x - 18$.

Solution: The G.C.D. is $x^2 - 3x + 3$.

Bézout's Theorem for Polynomials(裴蜀定理)

(Named after Etienne Bezout, 1730-1783, a French Mathematician.)

If $f(x)$ and $g(x)$ are two polynomials with the G.C.D. $h(x)$, then there exist two

other polynomials $F(x)$ and $G(x)$ such that $f(x)F(x) + g(x)G(x) = h(x)$.
 By suitable choice we can always have $\deg F(x) < \deg g(x)$ and $\deg G(x) < \deg f(x)$.
 For if $\deg F(x) > \deg g(x)$, then we can choose $\lambda(x)$ such that $F(x) - \lambda(x)g(x)$ is of degree less than $g(x)$, and obtain $f(x)[F(x) - \lambda(x)g(x)] + g(x)[G(x) + \lambda(x)f(x)] = 1$.
 Here we have $F'(x) = [F(x) - \lambda(x)g(x)]$ and $\deg F'(x) < \deg g(x)$, and

$$G'(x) = [G(x) + \lambda(x)f(x)] \text{ and } \deg G'(x) < \deg f(x).$$

In particular, if $f(x)$ and $g(x)$ are co-prime, then there exist polynomials $F(x)$ and $G(x)$ such that $f(x)F(x) + g(x)G(x) = 1$

Theorem: For any co-prime polynomials $f(x)$ and $g(x)$, if $F(x)$ and $G(x)$ are so chosen that $f(x)F(x) + g(x)G(x) = 1$, and $\deg F(x) < \deg g(x)$, $\deg G(x) < \deg f(x)$, then $F(x)$ and $G(x)$ are unique.

Proof: Suppose $F(x)$, $G(x)$ and $F_1(x)$, $G_1(x)$ satisfy the given conditions, that is $f(x)F(x) + g(x)G(x) = 1$ and $f(x)F_1(x) + g(x)G_1(x) = 1$, and $\deg F(x) < \deg g(x)$ and $\deg G(x) < \deg f(x)$, $\deg F_1(x) < \deg g(x)$ and $\deg G_1(x) < \deg f(x)$.
 then $[F(x) - F_1(x)]f(x) = [G_1(x) - G(x)]g(x)$.

Now $f(x)$ and $g(x)$ are co-prime, hence $f(x)$ is a factor of $[G_1(x) - G(x)]$.

However, $\deg [G_1(x) - G(x)] < \deg f(x)$.

This leads to contradictions.

The only possible case is that $G_1(x) = G(x)$, and similarly $F_1(x) = F(x)$.

That is, the existence of $F(x)$ and $G(x)$ are unique.

Example 28: Find the G.C.D. of $f(x) = x^4 + 3x^3 + x^2 - 2$ and $g(x) = x^4 - x^3 + x^2 + 2$,
 hence express the G.C.D. in the form $f(x)F(x) + g(x)G(x)$.

Solution: $x^2 + x + 1 = \left(\frac{1-x}{4}\right)f(x) + \left(\frac{x+3}{4}\right)g(x)$.

Example 29: Show that $f(x) = x^3 + x + 1$ and $g(x) = x^2 - x + 1$ are co-prime, and hence express 1 in the form $f(x)F(x) + g(x)G(x)$ for some polynomials $F(x)$ and $G(x)$.

Solution: $1 = (1-x)f(x) + (x^2)g(x)$.

Example 30: Find a polynomial $p(x)$ such that (x^2+1) is a factor of $p(x)$, and $x^3 + x^2 + 1$ is a factor of $p(x) + 1$.

Solution: $p(x) = (x^2 + 1)(x^2 + x - 1)$. (Note: The answer is not unique.)

Root of a Polynomial (多項式的根)

Given a polynomial $f(x)$. If $f(\alpha) = 0$, then α is called a zero or a root of the polynomial.

Fundamental Theorem of Algebra (代數基本定理/高斯定理)

Every non-constant single variable polynomial has at least one root, which may be real or complex.

Theorem: A polynomial of degree n ($n \geq 1$) has exactly n roots, though some of the roots may be repeated roots, and the roots may be real or complex.

Part 1: Using the Fundamental Theorem of algebra, it has at least one root, say r_1 .

Then $f(x) = (x - r_1)g(x)$, where $g(x)$ is a polynomial of degree $n-1$.

Suppose $n-1 \geq 1$, then again it has at least one root, say r_2 .

So, $f(x) = (x - r_1)(x - r_2)h(x)$

The process may be repeated until we obtain $f(x) = a_n(x - r_1)(x - r_2)\dots(x - r_n)$, where a_n is the coefficient of x_n in $f(x)$.

Hence $f(x)$ has n roots

Note: If a root occurs twice, it is called a double root. If it occurs three times, it is called a triple root. If it occurs m times, it is called a root of multiplicity m . (m 多重根)

Part 2: Suppose a polynomial $f(x)$ of degree n has m roots, where $m > n$.

Now, take any n roots $r_1, r_2, \dots, r_n$, we have

$$f(x) = a_n (x - r_1)(x - r_2) \dots (x - r_n).$$

Further, $f(x)$ is zero for at least one other root, say r_{n+1} .

$$\text{Hence } f(r_{n+1}) = a_n (r_{n+1} - r_1)(r_{n+1} - r_2) \dots (r_{n+1} - r_n)$$

We must have $a_n = 0$, and hence $f(x)$ is not a polynomial of degree n .

This is contradictory. Hence the assumption is wrong, and $f(x)$ has at most n roots.

Corollary: If $f(x)$ and $g(x)$ are polynomials of degree n , and $f(x_i) = g(x_i)$ with $x_1, x_2, \dots, x_m$ distinct numbers and $m > n$, then $f(x) \equiv g(x)$.

Proof: $f(x) - g(x)$ is a polynomial of degree less than or equal to n , but with m distinct roots, where $m > n$. Using the above Theorem, $f(x) - g(x) \equiv 0$, hence $f(x) \equiv g(x)$.
(This is used in **Method of Equating Coefficients of Polynomials**)

Example 31: Given that a, b and c are distinct real numbers, prove that

$$\begin{aligned} \text{(a)} \quad & \frac{(x-b)(x-c)}{(a-b)(a-c)} + \frac{(x-c)(x-a)}{(b-c)(b-a)} + \frac{(x-a)(x-b)}{(c-a)(c-b)} = 1 \\ \text{(b)} \quad & \frac{a^2(x-b)(x-c)}{(a-b)(a-c)} + \frac{b^2(x-c)(x-a)}{(b-c)(b-a)} + \frac{c^2(x-a)(x-b)}{(c-a)(c-b)} = x^2. \end{aligned}$$

Example 32: Given that $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ are $2n$ distinct real numbers, solve the following system of equations:

$$\begin{aligned} \frac{x_1}{a_1-b_1} + \frac{x_2}{a_1-b_2} + \dots + \frac{x_n}{a_1-b_n} &= 1, \\ \frac{x_1}{a_2-b_1} + \frac{x_2}{a_2-b_2} + \dots + \frac{x_n}{a_2-b_n} &= 1, \\ &\dots\dots\dots \\ \frac{x_1}{a_n-b_1} + \frac{x_2}{a_n-b_2} + \dots + \frac{x_n}{a_n-b_n} &= 1, \end{aligned}$$

Solution:

$$x_1 = - \frac{(b_1-a_1)(b_1-a_2) \dots (b_1-a_n)}{(b_1-b_2)(b_1-b_3) \dots (b_1-b_n)}$$

Similarly,

$$x_2 = - \frac{(b_2-a_1)(b_2-a_2) \dots (b_2-a_n)}{(b_2-b_1)(b_2-b_3) \dots (b_2-b_n)}$$

and

$$\dots\dots\dots x_n = - \frac{(b_n-a_1)(b_n-a_2) \dots (b_n-a_n)}{(b_n-b_2)(b_n-b_3) \dots (b_n-b_{n-1})}$$

Example 33: Prove that $(x+1)(x+2)(x+3)(x+4) + 1 = (x^2 + 5x + 5)^2$.

Example 34: Prove that $f(x) = (x-1)(x-2)(x-3) \dots (x-10) - 1$ is irreducible over the set of integers

Example 35: Let $f(x)$ be a polynomial of degree n and with integral coefficients. Prove that if there are at least $2n+1$ distinct integers m such that $|f(m)|$ is a prime number, then $f(x)$ is irreducible over the set of integers.

Example 36: (16th IMO 1974 No. 6) Let P be a non-constant polynomial with integer coefficients. If $n(P)$ is the number of distinct integers k such that $(P(k))^2 = 1$, prove that $n(P) - \deg(P) \leq 2$, where $\deg(P)$ denotes the degree of the polynomial P .

Method of Equating Coefficients of Polynomials

If two polynomials are identically equal to each other (i.e. they are equal to each other for all values of x), then their coefficients for corresponding powers of x can be equated. That is, if $ax^n + bx^{n-1} + \dots + k \equiv Ax^n + Bx^{n-1} + \dots + K$, then $a=A$, $b=B$, ..., $k=K$.

Example 36: Find the values of a and b so that $f(x) = x^4 + ax^3 + bx^2 - 12x + 9$ is the square of a polynomial.

Solution: The possible values of (a, b) are $(-4, 10)$ and $(4, -2)$

Example 38: Find the sum to n terms of the series $1^4 + 2^4 + 3^4 + \dots + n^4$

Solution: Finally, the sum = $\frac{n}{30}(n+1)(2n+1)(3n^2 + 3n - 1)$

Example 39: Find the sum to n terms of the series $1.2.3 + 2.3.4 + \dots$

Solution: $S(n) = \frac{n}{4}(n+1)(n+2)(n+3)$

Example 40: Find the sum to n terms of the series $1^3 + 3^3 + 5^3 + \dots$

Solution: Hence $1^3 + 3^3 + 5^3 + \dots \equiv n^2(2n^2 - 1)$